The empirical recurrence for OEIS A232340, recovered from its tail

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Abstract

OEIS A232340 records "Empirical recurrence of order 80" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1186$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 8$, so the recurrence is proved for every $n \geq 88$. Its last coefficient is $c_{80} = -192 \neq 0$, so no further tail satisfies anything shorter; any order-80 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A232340 is "Number of 6Xn 0..2 arrays with every 0 next to a 1 and every 1 next to a 2 horizontally or antidiagonally, with no adjacent elements equal.". It has offset 1 and begins

$$1, 288, 5338, 16338, 312276, 1047700, 20052758, 71916112, \ \ldots$$

The entry states:

> Empirical recurrence of order 80 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 07:19:43, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1186$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1186$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 80: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 8$ is the first whose minimal order is exactly the stated 80.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 2372$ exact terms from $n = 8$ on, it returns order 80 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{80} c_i\, a(n-i),$$

c_1	1	c_{21}	32657690	c_{41}	-191756283	c_{61}	-16526069
c_2	68	c_{22}	-8388711	c_{42}	172439832	c_{62}	-10551310
c_3	-38	c_{23}	42206234	c_{43}	135529678	c_{63}	-4540431
c_4	-206	c_{24}	-62290128	c_{44}	2937513	c_{64}	4326044
c_5	-731	c_{25}	-18042351	c_{45}	145609056	c_{65}	4594368
c_6	-3159	c_{26}	-60806289	c_{46}	-377685732	c_{66}	779974
c_7	6090	c_{27}	62211456	c_{47}	101178653	c_{67}	-360755
c_8	-6732	c_{28}	122684359	c_{48}	-249387442	c_{68}	-1459520
c_9	57720	c_{29}	25615293	c_{49}	327461437	c_{69}	-32297
c_{10}	-75059	c_{30}	40862540	c_{50}	-99308897	c_{70}	-100750
c_{11}	135513	c_{31}	-329354100	c_{51}	158048165	c_{71}	326180
c_{12}	-508067	c_{32}	102596011	c_{52}	-139004406	c_{72}	-10325
c_{13}	530133	c_{33}	-270266198	c_{53}	60495645	c_{73}	33122
c_{14}	-1067780	c_{34}	543860661	c_{54}	-40244497	c_{74}	-49082
c_{15}	2903631	c_{35}	-265850774	c_{55}	5114305	c_{75}	1717
c_{16}	-2376674	c_{36}	486651734	c_{56}	-32895376	c_{76}	-3937
c_{17}	5320896	c_{37}	-563687823	c_{57}	-10734025	c_{77}	4512
c_{18}	-11576825	c_{38}	317310071	c_{58}	28468330	c_{78}	-96
c_{19}	6666122	c_{39}	-465182584	c_{59}	19162380	c_{79}	192
c_{20}	-18131467	c_{40}	280926423	c_{60}	12552602	c_{80}	-192

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 88$, and it is the recurrence OEIS A232340 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 8$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{80} - c_1 x^{80-1} - \cdots - c_{80}$. Its constant term is $-c_{80} = 192$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 80 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 80, so $\deg q = 80$, and it is monic. It holds from some index t on. From $\max(t, 8)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 80, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 80, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -192.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-80 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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